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TUGAS 8 TEKNIK KOMPUTASI

1.) Integrasi Numerik untuk Menghitung Impuls Roket Uji Statik

Data yg diketahui :

t(s)	0	1	2	3	4	5	6
F(t)(kN)	0.0	3.2	5.1	6.0	5.4	3.1	0.4

$$\text{Impuls total : } J = \int_0^6 F(t) dt \quad m = 50 \text{ kg}$$

$$1 \text{ kNs} = 1000 \text{ Ns}$$

$$h = 1 \text{ s}$$

a.) Impuls total J dengan metode trapesium

$$J_T = \frac{h}{2} [F_0 + 2F_1 + 2F_2 + 2F_3 + 2F_4 + 2F_5 + F_6]$$

$$J_T = \frac{1}{2} [0 + 2(3.2) + 2(5.1) + 2(6.0) + 2(5.4) + 2(3.1) + 0.4]$$

$$J_T = \frac{1}{2} [6.4 + 10.2 + 12 + 10.8 + 6.2 + 0.4]$$

$$J_T = \frac{1}{2} (46)$$

$$J_T = 23.0 \text{ kNs}$$

b.) Impuls total J dengan Metode Simpson $\frac{1}{3}$

$$J_S = \frac{1}{3} [F_0 + 4F_1 + 2F_2 + 4F_3 + 2F_4 + 4F_5 + F_6]$$

$$J_S = \frac{1}{3} [0 + 4(3.2) + 2(5.1) + 4(6.0) + 2(5.4) + 4(3.1) + 0.4]$$

$$J_S = \frac{1}{3} [0 + 12.8 + 10.2 + 24 + 10.8 + 12.4 + 0.4]$$

$$J_S = \frac{1}{3} (70.6)$$

$$J_S = \frac{70.6}{3} = 23.53 \text{ kN.s}$$

c.) Impuls dari kN.s ke N.s

$$1 \text{ kN.s} = 1000 \text{ N.s}$$

$$\text{Metode Trapestum} \rightarrow J_T = 23.0 \times 1000 = 23000 \text{ N.s}$$

$$\text{Metode Simpson} \rightarrow J_S = 23.53 \times 1000 = 23.533 \text{ N.s}$$

d.) Δv ; $m = 50 \text{ kg}$

$$\Delta v = \frac{J}{m}$$

$$\text{Metode trapesium} \rightarrow \Delta v_T = \frac{23000}{50} = 460 \text{ m/s}$$

$$\text{Metode Simpson} \rightarrow \Delta v_S = \frac{23533}{50} = 470.67 \text{ m/s}$$

e.) Selisih relatif

$$\frac{|J_S - J_T|}{J_S} \times 100\%$$

$$= \frac{|23.533 - 23.000|}{23.533} \times 100\% = \frac{533}{23.533} \times 100\% = 0.02264 \times 100\% = 2.264\%$$

$$\text{Selisih relatif} = 2.264\%$$

2.) Turunan Numerik Untuk Perangkat Optik

Data Eksperimen :

x (mm)	-0.3	-0.2	-0.1	0.0	0.1	0.2	0.3
U(x) (μJ)	0.500	0.220	0.055	0.000	0.060	0.250	0.590

$$h = 0.1 \text{ mm}$$

a.) Selisih pusat untuk $\frac{du}{dx}$ di $x = 0.0 \text{ mm}$

$$\left(\frac{du}{dx}\right)_0 = \frac{U(0.1) - U(-0.1)}{2h}$$

$$\frac{du}{dx} = \frac{0.060 - 0.055}{2(0.1)} = \frac{0.005}{0.2} = 0.025 \text{ } \mu\text{J/mm}$$

b.) Gaya di $x = 0$ dari hasil a

$$F = -\frac{du}{dx}$$

$$F = -0.025$$

$$F(0) = -0.025 \text{ mN}$$

c.) Selisih pusat untuk menghitung $\frac{d^2u}{dx^2}$ di $x = 0.0 \text{ mm}$

$$\frac{d^2u}{dx^2} = \frac{U(0.1) - 2U(0) + U(-0.1)}{h^2}$$

$$= \frac{0.060 - 2(0) + 0.055}{(0.1)^2}$$

$$\frac{d^2u}{dx^2} = \frac{0.115}{0.01} = 11.5 \text{ } \mu\text{J/mm}^2$$

d.) Selisih maju utk menghitung $\frac{du}{dx}$ dan $\frac{d^2u}{dx^2}$ di $x = -0.3$ mm

$$\begin{aligned}\frac{du}{dx} &= \frac{u(-0.2) - u(-0.3)}{h} \\ &= \frac{0.220 - 0.500}{0.1} = \frac{-0.28}{0.1} = -2.8 \mu\text{J/mm}\end{aligned}$$

$$\begin{aligned}\frac{d^2u}{dx^2} &= \frac{u(-0.1) - 2u(-0.2) + u(-0.3)}{h^2} \\ &= \frac{0.055 - 0.44 + 0.5}{0.01} = \frac{0.115}{0.01} = 11.5 \mu\text{J/mm}^2\end{aligned}$$

e.) Selisih mundur di $x = 0.3$ mm

$$\frac{du}{dx} = \frac{u(0.3) - u(0.2)}{h}$$

$$\frac{du}{dx} = \frac{0.590 - 0.250}{0.1} = \frac{0.34}{0.1} = 3.4 \mu\text{J/mm}$$

$$\frac{d^2u}{dx^2} = \frac{u(0.3) - 2u(0.2) + u(0.1)}{h^2}$$

$$\frac{d^2u}{dx^2} = \frac{0.590 - 0.5 + 0.060}{0.01}$$

$$\frac{d^2u}{dx^2} = \frac{0.15}{0.01} = 15 \mu\text{J/mm}^2$$

f.) Komentar singkat apakah titik $x = 0.0$ mm dapat dianggap sbg titik stabil berdasarkan tanda $\frac{d^2u}{dx^2}$ di titik tsb

= Karena $\frac{d^2u}{dx^2}(0) = 11.5 > 0$ maka energi potensial memiliki titik minimum lokal pd $x = 0$. Jadi, $x = 0$ mm merupakan titik kesetimbangan stabil.

3.) Persamaan Differensial Gerak jatuh dengan Hambatan Kuadratik

$$\frac{dv}{dt} = g - \beta v^2 \quad \text{Dik: } g = 9.81 \text{ m/s}^2$$

$$\beta = 0.08 \text{ m}^{-1}$$

$$h = 0.5 \text{ s}$$

$$\frac{dv}{dt} = 9.81 - 0.08 v^2$$

$$v(0) = 0$$

a.) Metode Euler

$$v_{n+1} = v_n + hf(t_n, v_n)$$

$$v_{n+1} = v_n + 0.5(9.81 - 0.08 v_n^2)$$

n	t_n (s)	v_n (m/s)	$f(t_n, v_n)$ m/s ²
0	0.0	0	9.8100
1	0.5	4.905	7.8853
2	1.0	8.8476	3.5476
3	1.5	10.6219	0.7848
4	2.0	11.0138	-

$$t = 0 \rightarrow 0.5 \text{ s}$$

$$v_0 = 0$$

$$f(v_0) = 9.81 - 0.08(0)^2$$

$$f(v_0) = 9.81$$

$$v_1 = v_0 + 0.5 f(v_0)$$

$$v_1 = 0 + 0.5(9.81)$$

$$v_1 = 4.905$$

$$v_2 = v_1 + 0.5 f(v_1)$$

$$v_2 = 4.905 + 0.5(7.8853)$$

$$v_1 = 4.905$$

$$f(v_1) = 9.81 - 0.08(4.905)^2$$

$$v_2 = 4.905 + 3.9426$$

$$f(v_1) = 9.81 - 0.08(24.059)$$

$$v_2 = 8.8476$$

$$f(v_1) = 9.81 - 1.9247 = 7.8853$$

Fold

$$\bullet t = 1.0 \rightarrow 1.5 \text{ s}$$

$$V_2 = 8.8476$$

$$f(V_2) = 9.81 - 0.08(8.8476)^2$$

$$f(V_2) = 9.81 - 0.08(78.280)$$

$$f(V_2) = 9.81 - 6.2624$$

$$f(V_2) = 3.5476$$

$$V_3 = V_2 + 0.5 f(V_2)$$

$$V_3 = 8.8476 + 0.5(3.5476)$$

$$V_3 = 8.8476 + 1.7738$$

$$V_3 = 10.6214$$

$$\bullet t = 1.5 \rightarrow 2.0 \text{ s}$$

$$V_3 = 10.6214$$

$$f(V_3) = 9.81 - 0.08(10.6214)^2$$

$$f(V_3) = 9.81 - 9.0252$$

$$f(V_3) = 0.7848$$

$$V_4 = V_3 + 0.5 f(V_3)$$

$$V_4 = 10.6214 + 0.5(0.7848)$$

$$V_4 = 10.6214 + 0.3924$$

$$V_4 = 11.0138$$

b.) Metode Runge-Kutta orde 4

$$\bullet t = 0 \rightarrow 0.5$$

$$K_1 = f(V_0)$$

$$K_1 = 9.81 - 0.08(0)^2$$

$$K_1 = 9.81$$

$$K_2 = f(V_0 + \frac{h}{2} K_1)$$

$$K_2 = f(0 + 0.25(9.81))$$

$$K_2 = f(2.4525)$$

$$K_2 = 9.81 - 0.08(2.4525)^2$$

$$K_2 = 9.81 - 0.4812$$

$$K_2 = 9.3288$$

$$K_3 = f(V_0 + \frac{h}{2} K_2)$$

$$= f(0 + 0.25(9.3288))$$

$$= f(2.3322)$$

$$K_3 = 9.81 - 0.08(2.3322)^2$$

$$\rightarrow K_3 = 9.81 - 0.4351$$

$$K_3 = 9.3749$$

$$K_4 = f(V_0 + h(K_3))$$

$$= f(0 + 0.5(9.3749))$$

$$= f(4.6875)$$

$$K_4 = 9.81 - 0.08(4.6875)^2$$

$$K_4 = 9.81 - 1.7580$$

$$K_4 = 8.0520$$

$$V_1 = V_0 + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4)$$

$$V_1 = 0 + 0.08333(9.81 + 2(9.3288) + 2(9.3749) + 8.0520)$$

$$V_1 = 0.08333(55.2694)$$

$$V_1 = 4.6050 \text{ m/s}$$

$$t = 0.5 \rightarrow 1.0 \text{ s}$$

$$V_1 = 4.6058$$

$$K_1 = 9.81 - 0.08(4.6058)^2$$

$$K_1 = 8.1133$$

$$K_2 = f(4.6058 + 0.25(8.1133))$$

$$K_2 = f(6.6341)$$

$$K_2 = 6.28870$$

$$K_3 = f(4.6058 + 0.25(6.28870))$$

$$K_3 = f(6.17807)$$

$$K_3 = 6.75613$$

$$K_4 = f(4.6058 + 0.5(6.75613))$$

$$K_4 = f(7.98307)$$

$$K_4 = 4.71188$$

$$V_2 = 4.6058 + \frac{0.5}{6} (8.1133 + 2(6.28870) + 2(6.75613) + 4.71188)$$

$$V_2 = 4.6058 + 3.2429$$

$$V_2 = 7.8487 \text{ m/s}$$

3 :

$$t = 1.0 \rightarrow 1.5 \text{ s}$$

$$K_3 = f(7.8487 + 0.25(3.23))$$

5,

$$K_3 = f(7.8487 + 0.8075)$$

$$V_2 = 7.8487$$

$$K_3 = f(8.656)$$

$$K_3 = 9.81 - 0.08(8.656)^2$$

$$K_1 = 9.81 - 0.08(7.8487)^2$$

$$K_3 = 9.81 - 5.994$$

$$K_1 = 9.81 - 4.928$$

$$K_3 = 3.816$$

$$K_1 = 4.882$$

$$K_4 = f(7.8487 + 0.5(3.816))$$

$$K_2 = f(7.8487 + 0.25(4.882))$$

$$K_4 = f(9.7569)$$

$$K_2 = f(7.8487 + 1.2205)$$

$$K_4 = 2.19410$$

$$K_2 = f(9.0692)$$

$$K_2 = 9.81 - 0.08(9.0692)^2$$

$$V_3 = 7.8487 + \frac{0.5}{6} (4.882 + 2(3.23) + 2(3.816) + 2.19410)$$

$$K_2 = 9.81 - 6.58$$

$$K_2 = 3.23$$

$$V_3 = 7.8487 + 1.7640 = 9.6127 \text{ m/s}$$

Fold

$$t = 1.5 \rightarrow 2.0 \text{ s}$$

$$V_3 = 9.6127$$

$$K_1 = 9.81 - 0.08(9.6127)^2$$

$$K_1 = 9.81 - 7.39$$

$$K_1 = 2.42$$

$$K_2 = f(9.6127 + 0.5(2.41))$$

$$K_2 = f(9.6127 + 0.6025)$$

$$K_2 = f(10.2170)$$

$$K_2 = 9.81 - 0.08(10.2170)^2$$

$$K_2 = 9.81 - 8.350$$

$$K_2 = 1.46$$

$$K_3 = f(9.6127 - 0.25(1.46))$$

$$K_3 = f(9.6127 - 0.365)$$

$$K_3 = f(9.2477)$$

$$K_3 = 1.6204$$

$$K_4 = f(9.6127 - 0.08(1.6204))$$

$$K_4 = f(9.6127 - 0.1296)$$

$$K_4 = f(9.4831)$$

$$K_4 = 0.8454$$

$$V_4 = 9.6127 + \frac{0.5}{6} (2.42 + 2(1.46) + 2(1.6204) + 0.8454)$$

$$V_4 = 9.6127 + 0.8333(2.42 + 2.92 + 3.2408 + 0.8454)$$

$$V_4 = 9.6127 + 0.8333(9.42)$$

$$V_4 = 9.6127 + 0.7813$$

$$V_4 = 10.364 \text{ m/s}$$